

Projection matrix — and properties important for linear models

STAT 24510-30040

(Winter 2025)

Projection matrix is used extensively in statistics and machine learning, especially in linear regression models. In fact, we have applied the properties of projection matrices multiple times in this course.

Definition

An $n \times n$ matrix P is a projection matrix if P is symmetric and idempotent, that is,

$$P^T = P, \quad P^2 = P$$

(Recall P^T is the transpose matrix of P obtained by interchanging P 's rows into columns and columns into rows.)

Examples of projection matrices

- The $n \times n$ identity matrix I_n is a projection matrix.

This is because I_n is symmetric and idempotent.

$$I_n^T = I_n, \quad I_n^2 = I_n$$

- $P_X = X(X^T X)^{-1} X^T$, where X is $n \times p$ with $\text{rank}(X) = p$, with $p \leq n$.

P_X is a projection matrix.

- P_X is symmetric: $P_X = P_X^T$.

Proof. From the definition of matrix transpose, for any square matrices A, B and invertible C (exercise),

$$(AB)^T = B^T A^T, \quad (C^{-1})^T = (C^T)^{-1}$$

Using the above properties and the symmetry of $X^T X$,

$$P_X^T = (X(X^T X)^{-1} X^T)^T = (X^T)^T ((X^T X)^{-1})^T X^T = X(X^T X)^{-1} X^T = P_X$$

Thus P_X is symmetric.

□

- P_X is idempotent: $P_X^2 = P_X$.

Proof. To show the idempotency,

$$\begin{aligned} P_X^2 &= X(X^T X)^{-1} X^T X(X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} [X^T X(X^T X)^{-1}] X^T \\ &= X(X^T X)^{-1} I_n X^T = X(X^T X)^{-1} X^T = P_X \end{aligned}$$

□

This shows $P_X^2 = P_X$, therefore P_X is a projection matrix.

Remarks

The matrix $P_X = X(X^T X)^{-1} X^T$ is a very important matrix. It is the projection matrix to the column space of matrix X , critically used in linear models and other related applications.

- $P_v = v v^T$, $v \in \mathbb{R}^n$, $\|v\| = 1$.

P_v is a projection matrix.

The norm. $\|\cdot\|$ can be understood as the Euclidean norm induced by the inner product,

$$\|v\|^2 = \langle v, v \rangle = v^T v = \sum_{i=1}^n v_i^2, \quad \text{where } v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

We can check that, for any unit vector v with $\|v\| = 1$, the $n \times n$ matrix $v v^T$ is a projection matrix.

Proof. First, the matrix is symmetric,

$$(v v^T)^T = (v^T)^T v^T = v v^T$$

Since $v^T v = \|v\|^2 = 1$,

$$(v v^T)^2 = (v v^T) (v v^T) = v (v^T v) v^T = v v^T$$

Hence the idempotent property holds, Therefore $v v^T$ is a projection matrix.

□

- $J_n = \frac{1}{n} \mathbb{1} \mathbb{1}^T$, where $\mathbb{1} \in \mathbb{R}^n$ is the vector with entries all = 1.

J_n is a projection matrix.

This is a special case of $v v^T$ in the above example, with

$$v = \frac{1}{\sqrt{n}} \mathbb{1} = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{n} \\ \vdots \\ 1/\sqrt{n} \end{bmatrix}$$

We can check the unit-norm property,

$$\left\| \frac{1}{\sqrt{n}} \mathbb{1} \right\|^2 = \left(\frac{1}{\sqrt{n}} \mathbb{1} \right)^T \left(\frac{1}{\sqrt{n}} \mathbb{1} \right) = \sum_{i=1}^n \left(\frac{1}{\sqrt{n}} \right)^2 = 1$$

Basic properties of projection matrices

- If $n \times n$ matrix P is a projection matrix, then $I_n - P$ is also a projection matrix.

Proof. We verify the defining properties.

The matrix $I_n - P$ is symmetric,

$$(I_n - P)^T = I_n^T - P^T = I_n - P$$

To show the idempotent property,

$$(I_n - P)^2 = (I_n - P)(I_n - P) = I_n - I_n P - P I_n + P^2 = I_n - P - P + P = I_n - P$$

Therefore $I_n - P$ is a projection matrix. □

- If $n \times n$ matrix P is a projection matrix, then

$$P(I_n - P) = (I_n - P)P = 0_{n \times n}$$

Proof.

$$P(I_n - P) = P - P^2 = P - P = 0_{n \times n}$$

$$(I_n - P)P = P - P^2 = P - P = 0_{n \times n}$$

□

- Suppose P, Q are $n \times n$ projection matrices and

$$PQ = 0_{n \times n}$$

where $0_{n \times n}$ is the $n \times n$ matrix with all entries = 0. Then

$$P + Q$$

is also a projection matrix.

Proof. Again we verify the defining properties.

By the matrix transpose property,

$$(P + Q)^T = P^T + Q^T = P + Q$$

So $P + Q$ is symmetric. Note that

$$PQ = 0_{n \times n} \Rightarrow QP = Q^T P^T = (PQ)^T = 0_{n \times n}^T = 0_{n \times n}$$

Then the idempotent property follows,

$$(P + Q)^2 = (P + Q)(P + Q) = P^2 + PQ + QP + Q^2 = P^2 + Q^2 = P + Q$$

This proves that $P + Q$ is a projection matrix. □

Eigenvalues of projection matrices

- P is an $n \times n$ projection matrix. Then the eigenvalues of P are 1's and 0's.

Proof.

Linear algebra results on eigenvalues and eigenvectors are needed for this derivation.

Since P is symmetric, by symmetric matrix properties, P has n real eigenvalues $\lambda_i \in \mathbb{R}, i = 1, \dots, n$.

We may order the eigenvalues as

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n,$$

each with corresponding eigenvectors v_i ,

$$Pv_i = \lambda_i v_i, \quad i = 1, \dots, n.$$

In addition, by matrix and vector space properties, the n eigenvectors v_i can be chosen to be orthonormal, that is,

$$v_i^T v_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Let V be the matrix with v_i 's as column vectors,

$$V = [v_1 \ \dots \ v_n] \in \mathbb{R}^{n \times n}$$

By the orthonormal properties of the v_i 's,

$$V^T V = I_n$$

where I_n is the identity matrix. Thus V is an orthogonal matrix, with the nice property

$$V^{-1} = V^T$$

Let Λ be the diagonal matrix with λ_i as the i th diagonal entry. The eigenvalue-eigenvector relation

$$P[v_1 \ \dots \ v_n] = [\lambda_1 v_1 \ \dots \ \lambda_n v_n]$$

can be written as

$$PV = V\Lambda \Rightarrow P V V^{-1} = V \Lambda V^{-1} = V \Lambda V^T \Rightarrow P = V \Lambda V^T$$

The latter is the eigenvalue-eigenvector decomposition of matrix P .

By the projection matrix property $P^2 = P$ and $V^T V = I_n$,

$$P^2 = V \Lambda V^T V \Lambda V^T = V \Lambda^2 V^T = V \Lambda V^T = P$$

Multiplying $V \Lambda^2 V^T = V \Lambda V^T$ by V^T on the left and V on the right,

$$\Lambda^2 = V^T V \Lambda^2 V^T V = V^T V \Lambda V^T V = \Lambda$$

where $V^T V = I_n$ is used repeatedly. Since Λ is a diagonal matrix with λ_i 's on the diagonal,

$$\Lambda^2 = \Lambda \Rightarrow \lambda_i^2 = \lambda_i \Rightarrow \lambda_i = 0 \text{ or } 1, \quad i = 1, \dots, n.$$

This proves that the eigenvalues of P are 1's and 0's. □

- $Rank(P)$ = number of eigenvalues of P with eigenvalue = 1.

Proof. Since V is orthogonal thus full rank,

$$rank(P) = rank(\Lambda) = \#\{1\text{'s on the diagonal}\}$$

which is the number of eigenvalues of P with eigenvalue = 1.

□

- P is a projection matrix, then

$$trace(P) = rank(P)$$

Proof. We need to call in matrix trace and its properties.

Matrix trace of a square matrix is defined as the sum of diagonal elements,

$$trace(A) = tr(A) = \sum_{i=1}^n a_{ii}, \quad \text{where } A = [a_{ij}]_{n \times n}$$

From the definition, we can derive (exercise)

$$tr(A + B) = tr(A) + tr(B), \quad tr(AB) = tr(BA)$$

Then, by the eigenvalue-eigenvector decomposition of projection matrix P ,

$$tr(P) = tr(V\Lambda V^T) = tr(\Lambda V^T V) = tr(\Lambda I_n) = tr(\Lambda) = \sum_{i=1}^n \lambda_i = \text{number of } \lambda_i = 1.$$

where λ_i 's are eigenvalues of P , with possible values 1 and 0, as shown before.

□

Spectrum decomposition of projection matrices

Consider an $n \times n$ project matrix P with $Rank(P) = k$, and eigenvalue-eigenvector decomposition

$$P = V\Lambda V^T$$

where the columns of V are orthonormal eigenvectors. Order the eigenvectors $v_1, \dots, v_p$ such that

$$Pv_i = \begin{cases} v_i, & i = 1, \dots, k; \\ 0, & i = k+1, \dots, n. \end{cases}$$

Hence Λ is diagonal with the first r diagonal entries =1 followed by $n-r$ diagonal entries = 0.

$$V\Lambda V^T = [v_1 \ v_2 \ \dots \ v_n] \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 0 \\ & & & & \ddots \\ & & & & & 0 \end{bmatrix} \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_r^T \\ \vdots \\ v_n^T \end{bmatrix} = [v_1 \ v_2 \ \dots \ v_n] \begin{bmatrix} v_1^T \\ \vdots \\ v_r^T \\ 0 \\ \vdots \\ 0 \end{bmatrix} = v_1 v_1^T + \dots + v_r v_r^T$$

Therefore P can be expressed as the sum of orthogonal rank-1 projection matrices:

$$P = v_1 v_1^T + \dots + v_r v_r^T$$

An application

As an application, below we show an important projection matrix property which has been applied on many occasions in our study of linear regression models.

Corollary

If P is a projection matrix of rank r and $Z = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix} \sim N_n(0_n, I_n)$, that is, Z_i 's are *i.i.d.* $\sim N(0, 1)$,

then

$$\|PZ\|^2 \sim \chi_n^2$$

Proof. Use the previously derived eigenvalue-eigenvector decomposition, we may write

$$P = V\Lambda V^T$$

where the columns of V are orthonormal eigenvectors of P , and Λ is a diagonal matrix with the eigenvalues of P on the diagonal.

Since $rank(P) = r$, and since the eigenvalues of P are 1's and 0's, we can order the orthonormal eigenvectors in V so that the diagonal matrix Λ can be chosen with each of the first r diagonal entries =1 followed by $n-r$ diagonal entries = 0.

$$\|PZ\|^2 = (PZ)^T(PZ) = Z^T P^T P Z = Z^T P^2 Z = Z^T P Z = Z^T V \Lambda V^T Z = (V^T Z)^T \Lambda (V^T Z)$$

Define

$$V^T Z = Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \in \mathbb{R}^n$$

Then

$$\|PZ\|^2 = (V^T Z)^T \Lambda (V^T Z) = Y^T \Lambda Y$$

As an orthogonal linear transformation of $Z \sim N_n(0_n, I_n)$, $Y = V^T Z$ is also of n -variate normal, with mean vector

$$\mathbb{E}(Y) = V^T \mathbb{E}(Z) = 0_n$$

and covariance matrix

$$Cov(Y) = Cov(V^T Z) = V^T Cov(Z) (V^T)^T = V^T I_n V = V^T V = I_n$$

Therefore,

$$Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \sim N_n(0_n, I_n) \quad \Rightarrow \quad Y_1, \dots, Y_n \quad i.i.d. \quad \sim N(0, 1)$$

Since the top r diagonal entries of Λ are 1's and the rest are 0's,

$$\|PZ\|^2 = (V^T Z)^T \Lambda (V^T Z) = Y^T \Lambda Y = Y_1^2 + \dots + Y_r^2 \sim \chi_r^2$$

□